```
!pip install openai-whisper noisereduce pandas openpyxl
```

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```

```
import whisper
import pandas as pd
import numpy as np
import noisereduce as nr
import librosa
import soundfile as sf
import torch
import os

# 1. Clear memory and load a light model
torch.cuda.empty_cache()
model = whisper.load_model("turbo")
# We use 'turbo' for speed and memory efficiency

# 2. Define the Preprocessing function
def preprocess_audio(audio_path):
    data, rate = librosa.load(audio_path, sr=16000)
    reduced_noise = nr.reduce_noise(y=data, sr=rate)
    clean_path = "temp_clean.wav"
    sf.write(clean_path, reduced_noise, rate)
    return clean_path

# 3. Define the Transcription function
def transcribe_eit(audio_path):
    clean_file = preprocess_audio(audio_path)

    result = model.transcribe(clean_file, language="es", verbose=False)
    return result['text'].strip()

print("Functions defined and model loaded successfully!")
```

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Saving 038010_EIT-2A.mp3 to 038010_EIT-2A.mp3
Saving 038011_EIT-1A.mp3 to 038011_EIT-1A.mp3
```

```
data_for_excel = []

for filename in uploaded.keys():
    print(f"Processing: {filename}")
    transcription = transcribe_eit(filename)

    # I created a placeholder for 'Target Sentence' which I will fill
    # from the AutoEIT Sample spreadsheet later.
    data_for_excel.append({
        "File Name": filename,
        "Raw ASR Output": transcription,
        "Final Transcript (Human Corrected)": "",
        "Observations (Noise/Hesitation)": ""
    })

df = pd.DataFrame(data_for_excel)
df.to_excel("AutoEIT_Draft_Results.xlsx", index=False)
```

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```
Processing: 038015_EIT-1A.mp3
100%|■■■■■■■■■■| 52937/52937 [00:20<00:00, 2611.24frames/s]
Processing: 038012_EIT-2A.mp3
100%|■■■■■■■■■■| 112554/112554 [01:01<00:00, 1829.76frames/s]
Processing: 038010_EIT-2A.mp3
100%|■■■■■■■■■■| 54963/54963 [00:40<00:00, 1365.93frames/s]
Processing: 038011_EIT-1A.mp3
100%|■■■■■■■■■■| 54703/54703 [00:36<00:00, 1495.24frames/s]
```

```
: re

ting an Excel writer object
d.ExcelWriter("AutoEIT_Transcriptions_by_Participant.xlsx") as writer:

lename in uploaded.keys():

w_text = df.loc[df['File Name'] == filename, 'Raw ASR Output'].values[0]

Split into sentences (splitting by . ! or ?)
ntences = re.split(r'(?<=[.!?]) +', raw_text)


rticipant_data = []
```

```
· i, sent in enumerate(sentences):
participant_data.append({
    "Sentence #": i + 1,
    "Target Sentence (Paste here)": "",
    "ASR Raw Output": sent,
    "Final Transcript (Human Corrected)": sent,
    "Observations (Noise/Hesitation)": ""
})

Convert to a temporary DataFrame
np_df = pd.DataFrame(participant_data)

Write to a specific sheet named after the file (shortened to 31 chars for Excel)
eet_name = filename[:30]
np_df.to_excel(writer, sheet_name=sheet_name, index=False)

Multi-tab Excel generated! Download 'AutoEIT_Transcriptions_by_Participant.xlsx'.")
```

```
Multi-tab Excel generated! Download 'AutoEIT_Transcriptions_by_Participant.xlsx'.
```

Start coding or generate with AI.

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